

frequency response of the op-amp and is usually expressed as gain-bandwidth product or unity-gain crossover frequency. This parameter gives the small signal bandwidth. That is, signal amplitude and frequency are such that the rate of change of output does not exceed the slew rate specification. Fig. 1 shows the frequency response of LF156.

Slew rate is the rate of change of output with time under large signal conditions. It is determined by applying a step input and monitoring the output. It decides the large signal bandwidth of the op-amp.

Fig. 2 shows the small-signal and large-signal pulse response of LF156. Increased rise time in the case of large-signal pulse response due to slew rate limitation is evident in Fig. 2(b).

Common-mode rejection ratio (CMRR) is the ability of the op-amp to amplify differential signals and reject common-mode signals. This parameter is particularly important in data acquisition applications where it is desirable to amplify low-level differential signals riding on high-amplitude, common-mode signals. CMRR of 80 to 90 dB is common in general-purpose op-amps. Instrumentation op-amps offer CMRR in excess of 100 dB. AD524, an instrumentation op-amp, has CMRR of greater than 120 dB.

Input impedance is the impedance across input terminals of the op-amp. Usually expressed in terms of resistance only, it is different for inverting and non-inverting amplifier circuits. For inverting amplifier circuits it equals the input resistance connected from the source of signal to the inverting input of the op-amp, and for non-inverting amplifier circuits it equals the product of loop gain and

specified input impedance.

Types of op-amps

As mentioned earlier, the first step in choosing the right op-amp for a given application is to know and appreciate the category that the desired op-amp is likely to belong to. This helps in quick identification of the device as the operational amplifier manufacturers in their databooks often categorise their devices into one or more of the types depending upon the range of devices offered by them.

General-purpose op-amps are those op-amps which have moderate or reasonably good values of various key parameters. None of the major performance parameters is either exceptionally good or very poor. Among the broad category of general-purpose op-amps, those with better overall performance specifications are referred to as high-performance, general-purpose op-amps. While LM741/741C is a general-purpose op-amp, LF156/256/356 with its relatively better bandwidth, slew rate, input impedance and noise specifications may be categorised as a high-performance, general-purpose op-amp.

High-speed and high-bandwidth op-amps have relatively much higher slew rate and bandwidth specifications. These specifications are usually a couple of orders of magnitude higher than general-purpose op-amps. For example, the slew rate and gain-bandwidth values of AD829, which is a high-speed, high-bandwidth op-amp from Analog Devices, are 230 V/ μ s and 120 MHz, respectively. You can see the difference for yourself when you compare these values with the corresponding values of LM741C and LF356. The slew rates of LM741C

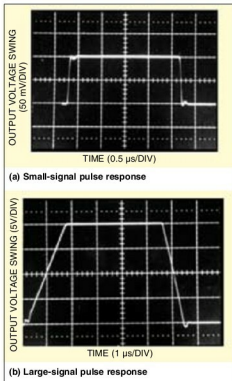


Fig. 2: Signal pulse response of LF156

and LF356 are 0.5V/ μ s and 12V/ μ s, respectively. Gain bandwidths are 1 MHz (for LM741C) and 5 MHz (for LF356).

Op-amp comparators are specially designed to operate in comparator applications. These respond relatively much faster—within a few nanoseconds to tens of nanoseconds. Their outputs are usually compatible to common logic families. One such op-amp comparator is LM 119/219/319, which is a dual comparator. Response time is specified to be 80 ns and its open-collector output with the recommended pull-up resistor can drive RTL, DTL and TTL family devices.

Norton op-amps, also known as current differencing op-amps, have a different internal architecture that makes them particularly suitable for single-supply operation. While we talk about differential input voltage in the case of conventional op-amps, it is differential input current for Norton op-amps. Most general-purpose op-amp applications can be realised using Norton op-amps without significantly compromising the performance characteristics. This, coupled with low cost and single supply operation, makes

Different categories of op-amps range from general-purpose op-amps offering moderate values of all key parameters to specialised types including high-speed op-amps, op-amp comparators, instrumentation op-amps, isolation op-amps and Norton op-amps.